IGI LIMS

An illustrated tour of the specimen tracking system, as built to date

PIRL, UNC · October 28, 2026 · status: prototype, specification in flight

1. What it is

IGI LIMS is a chain-of-custody database and web interface for the biological specimens collected in the IGI cancer clinical trial. It records one row per tube. Every tube stores the tube it was made from, so the path from a received specimen, through fractionation, cryopreservation, enrichment and extraction, to the sequencing or mass-spectrometry data file is a chain of parent links that can be walked in either direction. It runs as a single Flask application on SQLite or PostgreSQL, sized for a lab of about two dozen users, on a workstation, a GCP virtual machine, or Vercel with Supabase as the public demonstration does.

The screenshots in this document were taken from a throwaway demonstration database. The participant, tubes, users and lists shown are invented. Red numbered arrows mark the features explained under each picture.

2. The tube identifier

Identifiers follow IGI Sample Nomenclature v2.1. They are composed by the database from the tube's attributes and are never typed by hand. The format is defined in one configuration list, and the previous v2 format is kept in the code history as a fallback.

```
IGI-CC-PNNN-UNCNNN-YYMMMDD-SS-FFF-MMM-AA-OO(O)-YYMMMDD (51 or 52 characters)
```

Segment	Meaning	Source and validation
IGI	Project constant	literal
CC	Cohort	participant; active code in the cohort list (MP, DG, AM, TN, NC)
PNNN	IGI patient ID	participant; issued in sequence (P001, P002, ...), never reused
UNCNNN	UNC ID	participant; UNC + the 3-digit UNC number
YYMMMDD	Received date	collection; e.g. 26OCT06, shared by every tube from the collection
SS	Specimen	collection; active code (PF, PB, BM, TU, LN, NM)
FFF	Fraction	this tube; active code (PFS, PFM, PLA, TUM, IMM, CD8, NAP, ...)
MMM	Material	this tube; active code (FRZ, CPU, SNF, CUL, DNA, HMD, RNA, PEP, ...)
AA	Aliquot	01 to 99, counted within the first eight segments, never reused
OO(O)	Operator	this tube; initials of an active user, 2 or 3 letters, unique
YYMMMDD	Processing date	this tube; the date it was made

The walkthrough that follows produces, for example, `IGI-MP-P001-UNC316-26OCT06-PF-PFM-CPU-01-JD-26OCT06`: cohort MP, participant P001 (UNC316), received 6 October 2026, pleural fluid, PFMC, cryopreserved, aliquot 01, made by JD the same day. The specimen as received is a *collection* record, not a tube. Tubes made straight from it have no parent; every later tube keeps the first six segments of its parent and takes its own fraction, material, aliquot, operator and processing date. The label prints the ID in three rows of at most 18 characters, broken at hyphens, beside a Data Matrix barcode of the full ID.

3. What the system enforces

- **Controlled vocabularies.** Every code used in IDs, forms and metadata lives in one code-list table managed on the Codes page. Codes are retired rather than deleted, so historical records keep their meaning.
- **No free text except Notes.** Operator, SOP, storage location, custody parties and purposes, and every metadata vocabulary (kits, lots, buffers, media, antibodies, instruments) are dropdowns. The server rejects values that are not active list entries. The only remaining free-text fields are machine identifiers (data file URI, run ID, checksum).
- **Derivation rules.** A rule table decides which fractions a specimen type may have, which fraction and material transitions are allowed, and which materials can be made straight from a received specimen. A processing date before the received date is also caught. A violation blocks creation and offers an override with a written justification instead.
- **Label values are read-only.** The received date, operator and processing date are printed on the label, so they cannot be edited once a tube exists. A mistake is fixed with a correction: the right value and the reason are recorded beside the original, and the ID and label stay as printed.
- **Changes are flagged, not queued.** Edits, deletions, restores and rule overrides apply immediately, with the operator's reason, and are flagged in the audit log and on the tube page. The audit log has a changes-only view for review.
- **Soft deletes and a full audit log.** Deleting a tube keeps the record and its aliquot number, and is refused while the tube has active children. Every write is recorded with who, when, and before and after values.

4. Walkthrough: signing in and finding your way

4.1 Sign in

Every page requires a login. Accounts are created by an admin from the Users page or the command line.

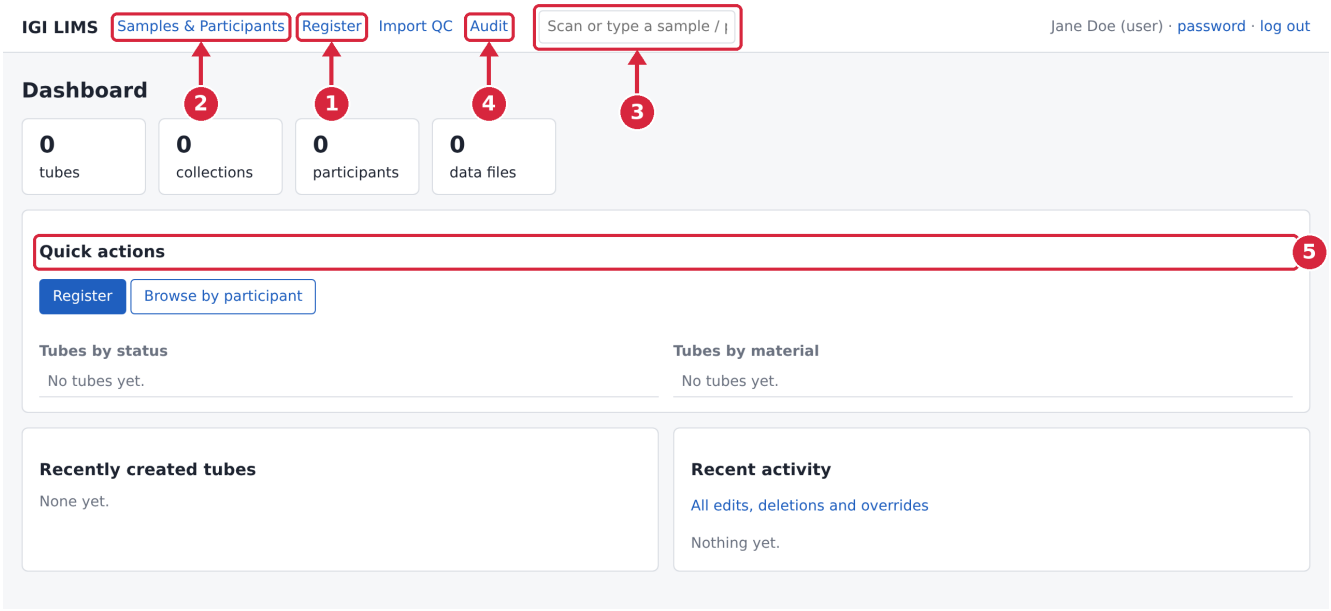


1 Username (the account name, not the display name)

2 Sign in

4.2 The dashboard

The dashboard after signing in. Nothing on it is editable; it links to the three working areas.

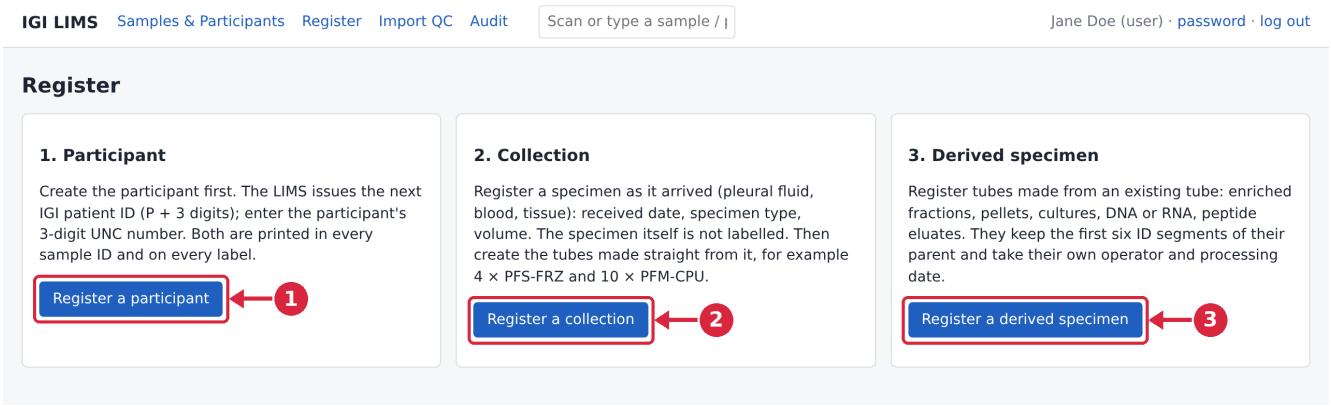


1. Register: the only place records are created
2. Samples & Participants: read-only browsing
3. Jump box: scan or paste a tube ID to open it
4. Audit: every write, with edits and overrides flagged
5. Quick actions, counts, recent tubes and recent activity

4.3 The Register page

All record creation starts here. The three cards are the three kinds of record a lab member creates, and the order matches the order of events: a participant enrols, a specimen arrives, tubes are made from it.

The Register page offers exactly three actions, in the order they happen in the lab.

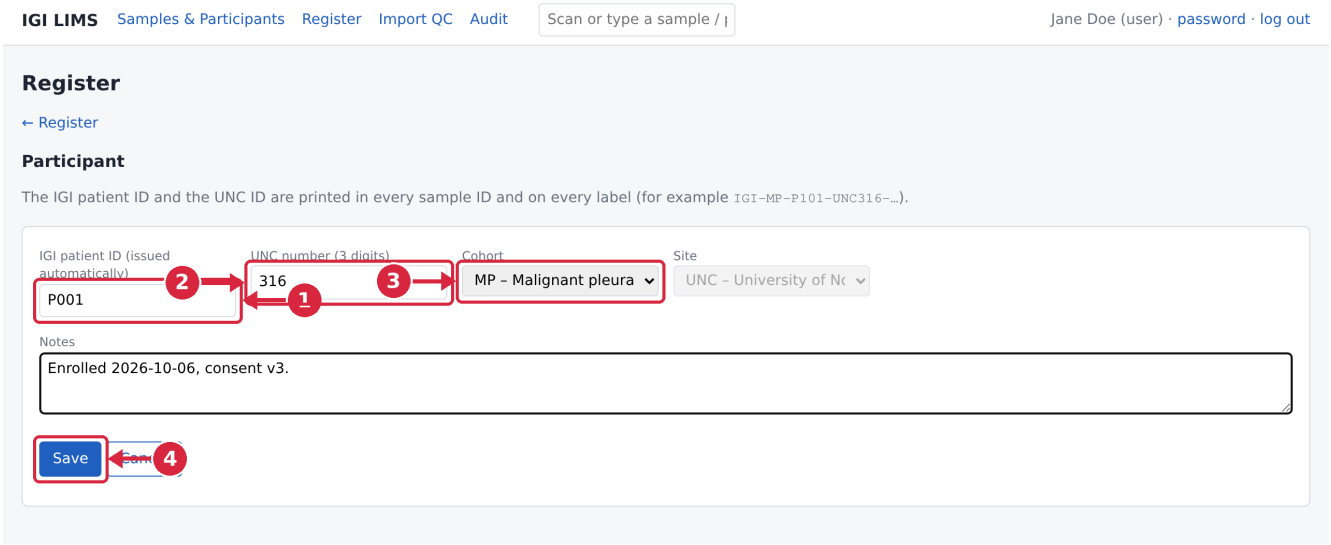


1. Step 1: a person enrolled in the trial
2. Step 2: a specimen as received, then the tubes made from it
3. Step 3: a tube made from an existing tube

5. Walkthrough: a participant and a collection

5.1 Register a participant

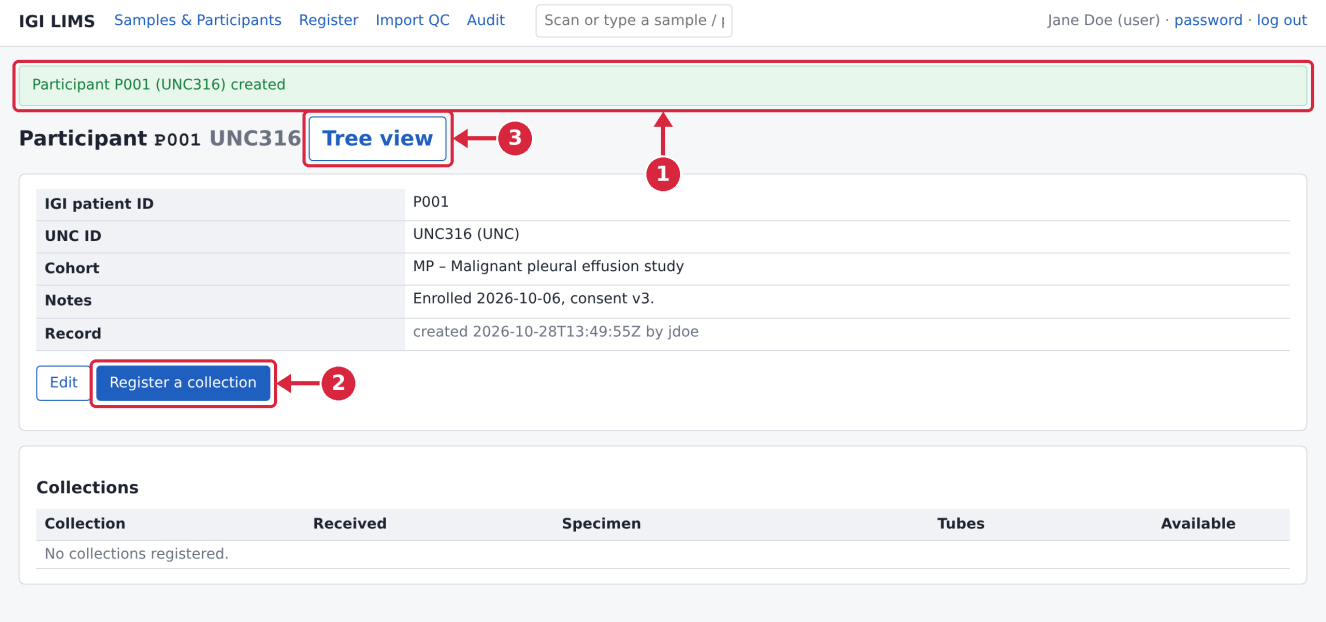
Registering a participant. The IGI patient ID and the UNC ID both appear in every sample ID and on every label.



1. IGI patient ID: the next number is issued (admins may enter one)
2. UNC number, 3 digits: printed as UNC316 in every ID
3. Cohort, from the cohort code list
4. Save

5.2 The participant record

The participant record. It lists every collection and links to the tree.



1 Confirmation with the issued IGI patient ID

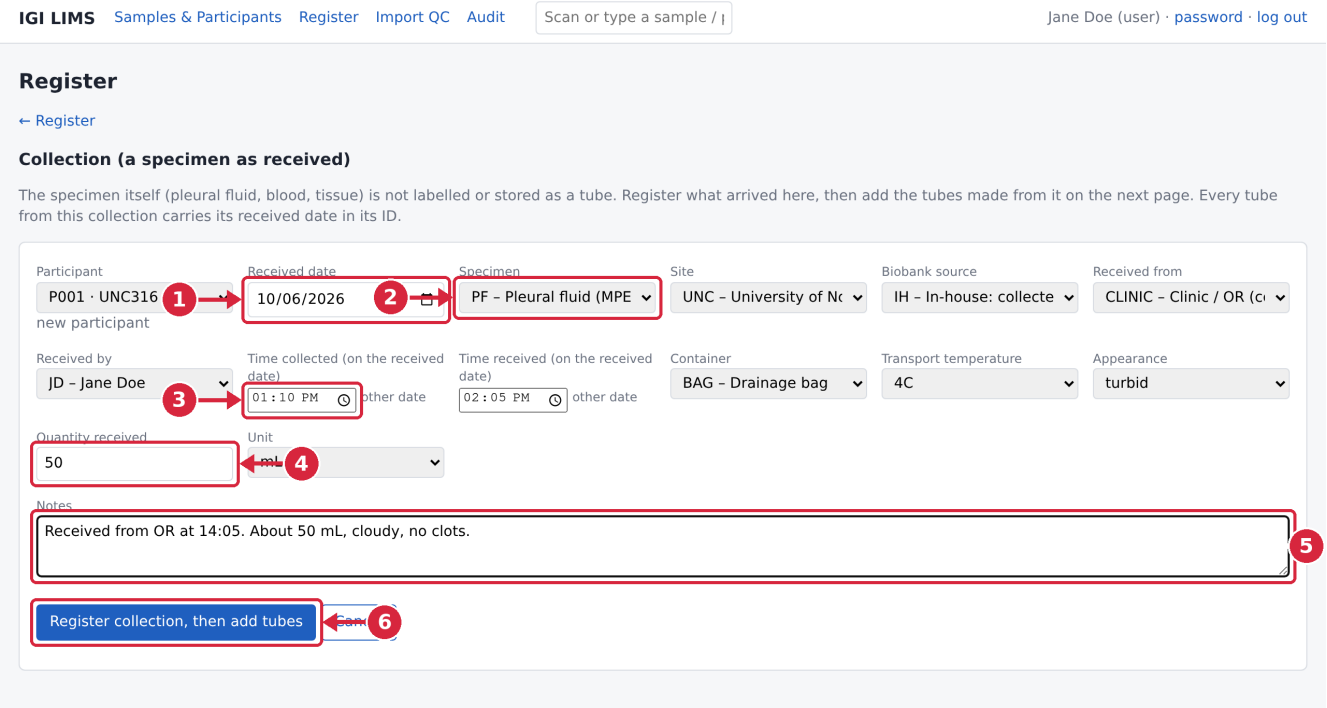
2 Go straight to registering this participant's first collection

3 Opens the tree for this participant

5.3 Register a collection

A collection is the specimen as it arrived: pleural fluid, blood or tissue. It is recorded once, with its received date, volume, container and times, and it is not labelled or stored as a tube. Its received date appears in the ID of every tube made from it.

Registering a collection: the pleural fluid as it arrived. All fields are required; NA is an explicit choice where a field does not apply.



1 Received date: printed in every tube ID from this collection (26OCT06)

2 Specimen: pleural fluid, blood, tissue ...

3 Times are entered as times; the date is the received date

4 What arrived: the fluid itself is never labelled or stored as a tube

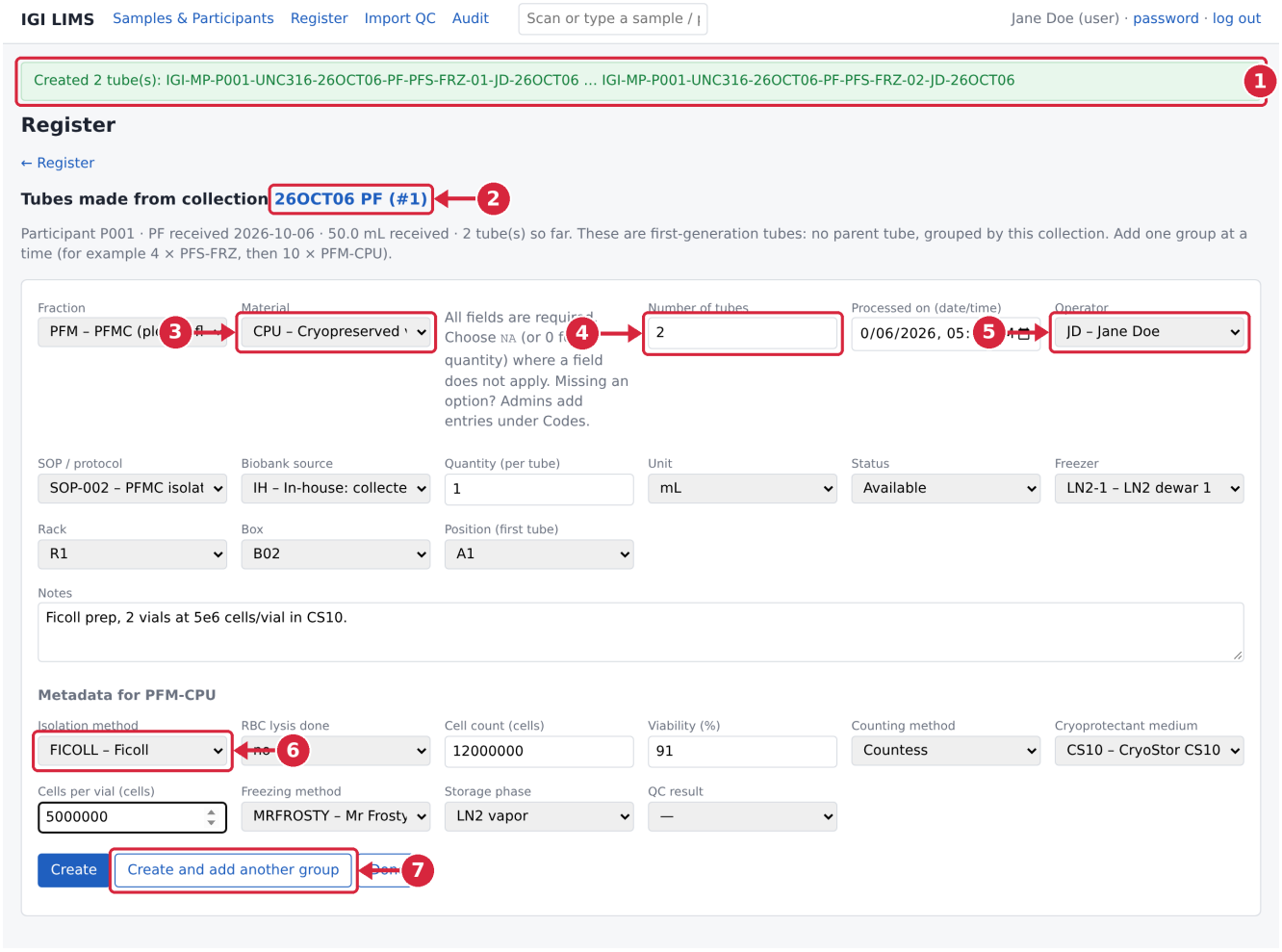
5 Notes: the only free-text field in the system

6 Register, then add the tubes made from it

5.4 Add the tubes made from it

Registering the collection leads straight to this form. Tubes made directly from the specimen are first-generation tubes. Here two tubes of cell-free supernatant were created first, and two PFMC cryovials are about to be. The metadata block changes with the fraction and material chosen.

Tubes made straight from the collection are first-generation tubes: they have no parent tube. Add one group at a time.



1 The previous group (2 x PFS-FRZ supernatant) was created

2 The collection these tubes come from

3 Fraction and material of this group: PFMC, cryopreserved

4 Two tubes get consecutive aliquot numbers

5 Operator: printed in the ID

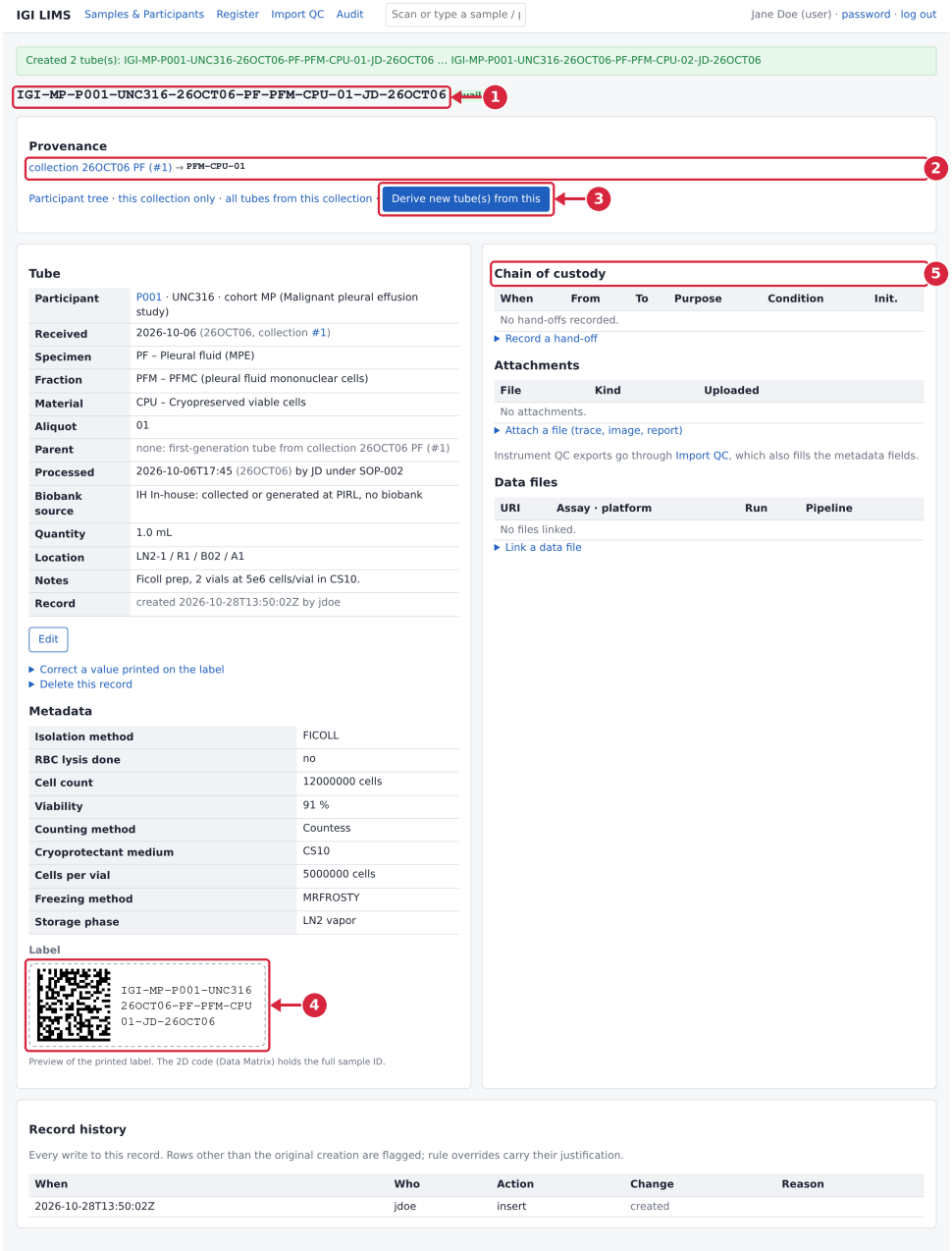
6 Metadata fields for this fraction and material

7 Create and stay here for the next group

5.5 The tube page

Every tube has one page: provenance from the collection, the tube's own fields, metadata, the label with its barcode, custody events, attachments, linked data files and its full change history.

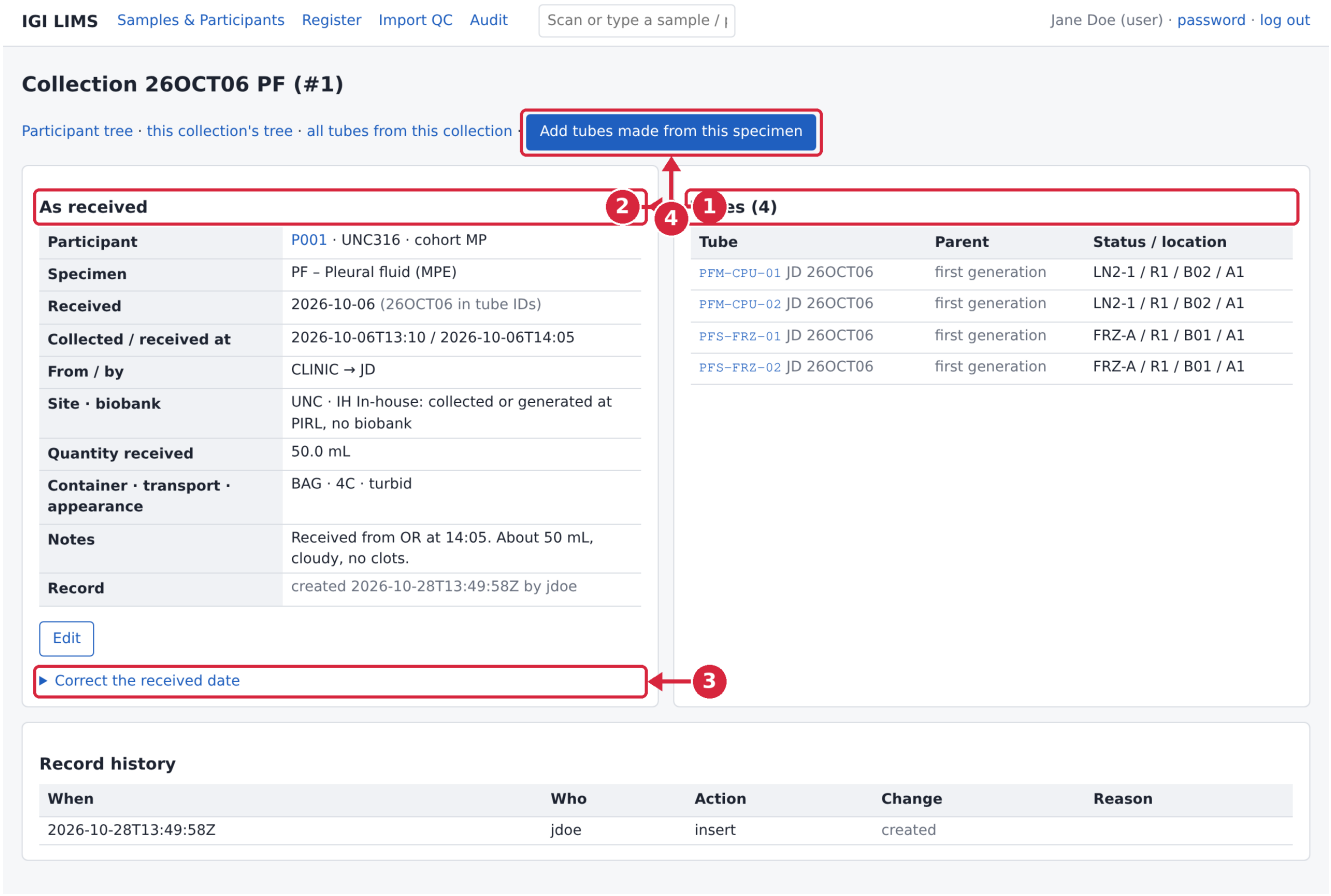
The tube page for the first PFMC cryovial.



1. The issued ID, 51 characters
2. Provenance: collection, then this tube
3. Derive children from this tube
4. The label: three rows broken at hyphens, plus a Data Matrix of the ID
5. Append-only hand-off log

5.6 The collection page

The collection page.



1 The collection record: what arrived, when and from whom

2 Every tube from this collection, first generation and derived

3 The received date is in every ID, so it is corrected, never edited

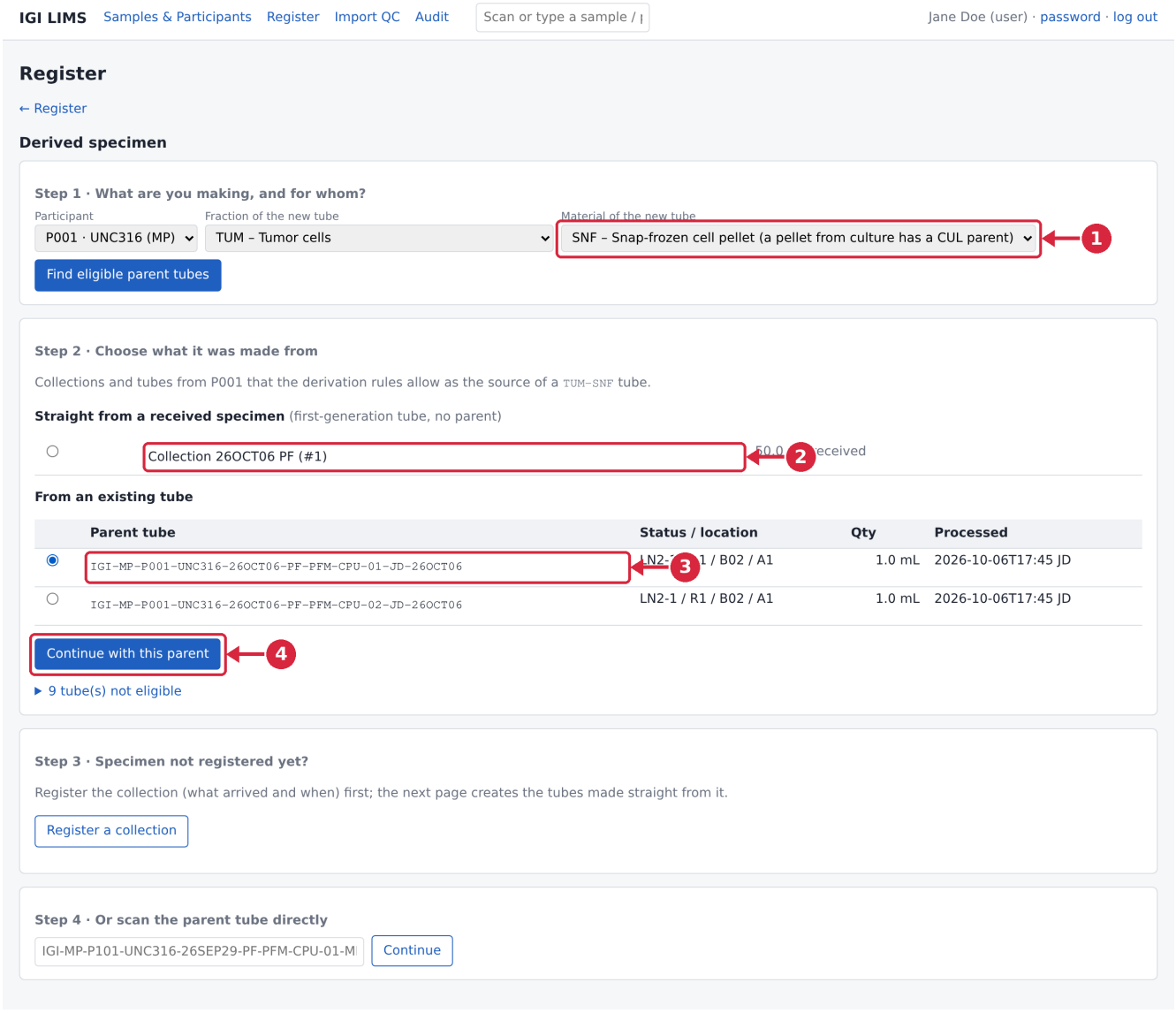
4 Add more first-generation tubes later

6. Walkthrough: deriving tubes

A derived tube is any tube made from another: an enriched fraction from a cryovial, a pellet, DNA from a pellet. The registration flow asks what is being made first and then offers only the sources the rules allow, so the operator never has to work out eligibility by hand.

6.1 Choose what to make, then choose what it was made from

Registering a derived specimen starts from the product, not the parent. Here: tumour-cell pellets, snap frozen.

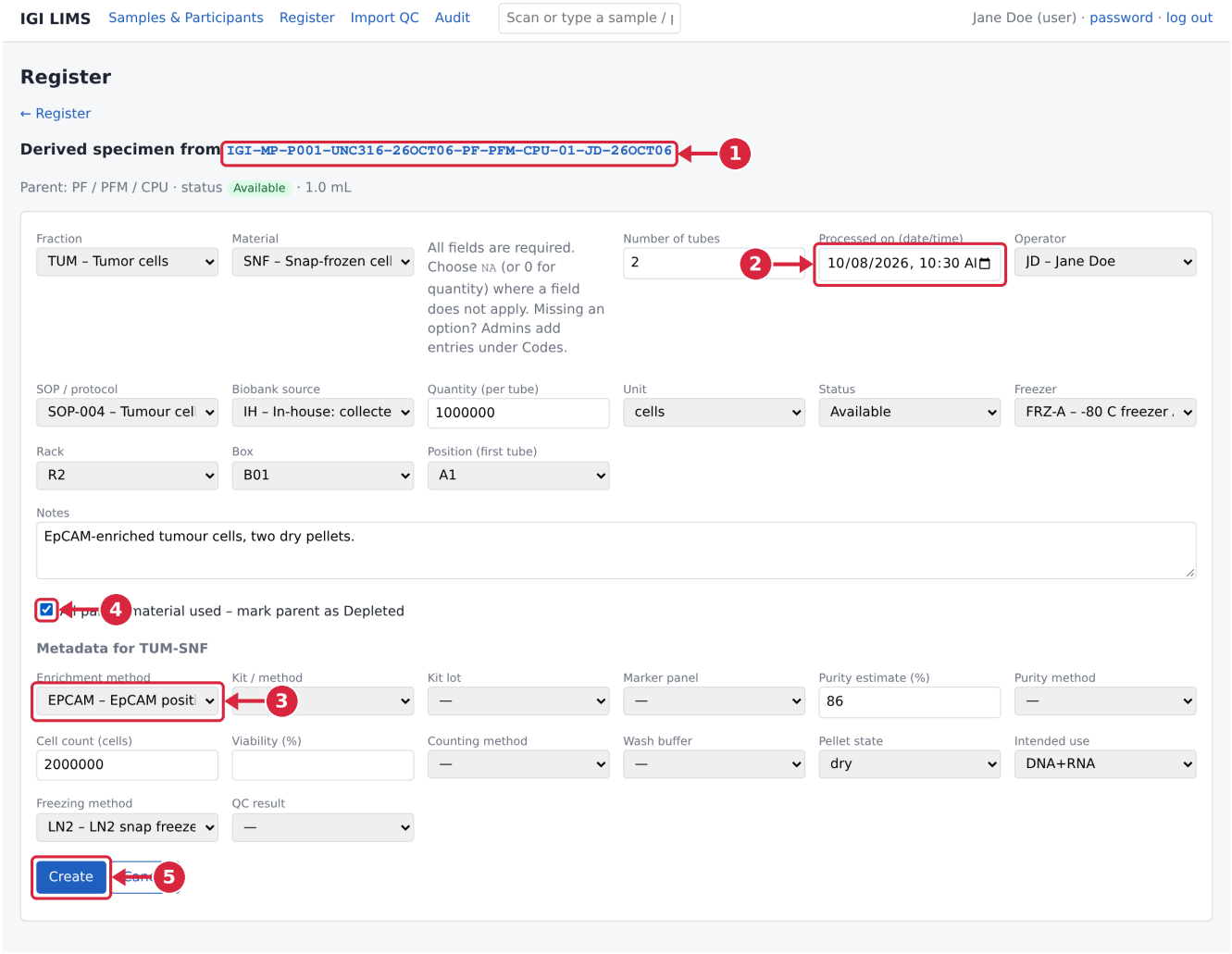


1. Say what the new tube will be: participant, fraction and material
2. Straight from a collection is offered too, when the rules allow it
3. Tubes the derivation rules accept as inputs
4. Continue with the selected parent

6.2 Fill in the new tubes

Here two tumour-cell pellets are made from the first cryovial, which is used up in the process.

The derived-tube form. The new tubes keep the first six ID segments of the parent and take their own operator and processing date.



1 The parent tube

2 Processing date: becomes the last ID segment (26OCT08)

3 Metadata for TUM and SNF: enrichment, purity, pellet state

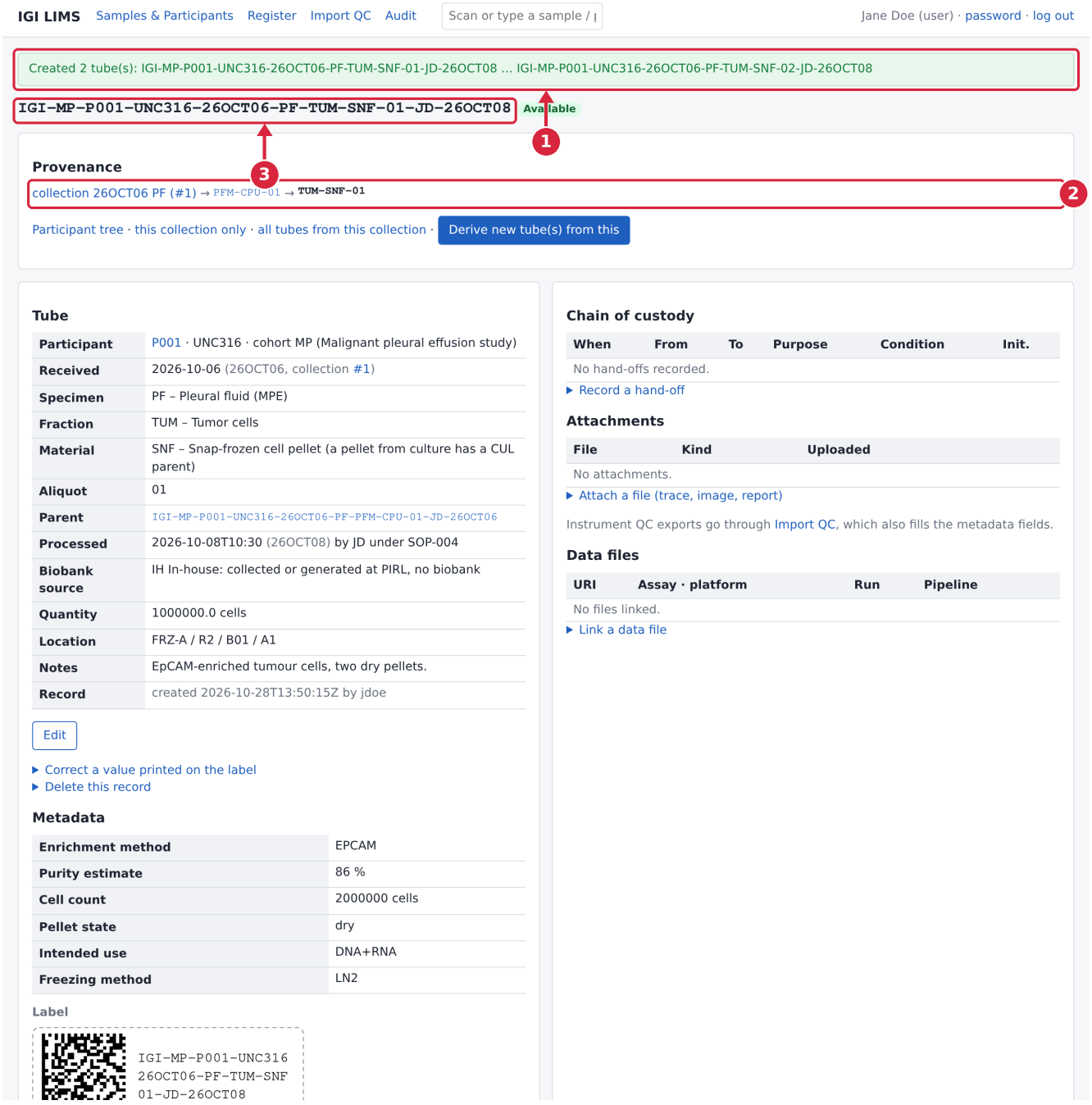
4 Marks the parent Depleted in the same step

5 Create

6.3 The result

The first pellet is IGI-MP-P001-UNC316-26OCT06-PF-TUM-SNF-01-JD-26OCT08. The cryovial it came from is now Depleted and one click away.

The first pellet. Its parent cryovial is now Depleted.



1 Two tubes created: ...-01 and ...-02

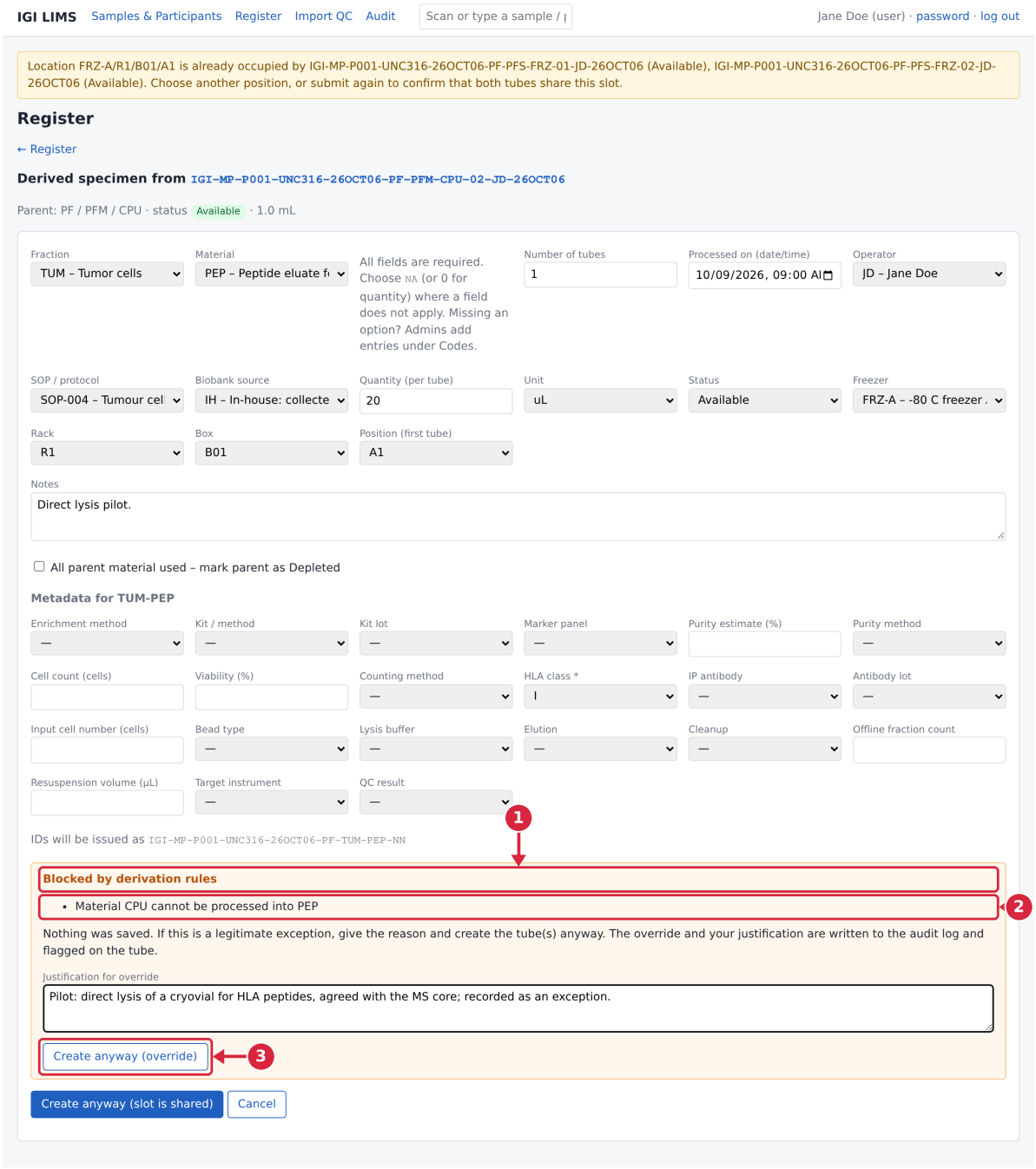
2 Collection, cryovial, pellet: every step is one click away

3 Same first six segments as the parent; own fraction, material, operator and date

6.4 When the rules say no

Here the operator tried to make a peptide eluate directly from a cryovial; the rules expect cells to be pelleted first. The form refuses, lists the reason, and allows an override only with a written justification.

Trying to make a peptide eluate straight from a cryovial. The derivation rules refuse; the operator may override with a written justification.



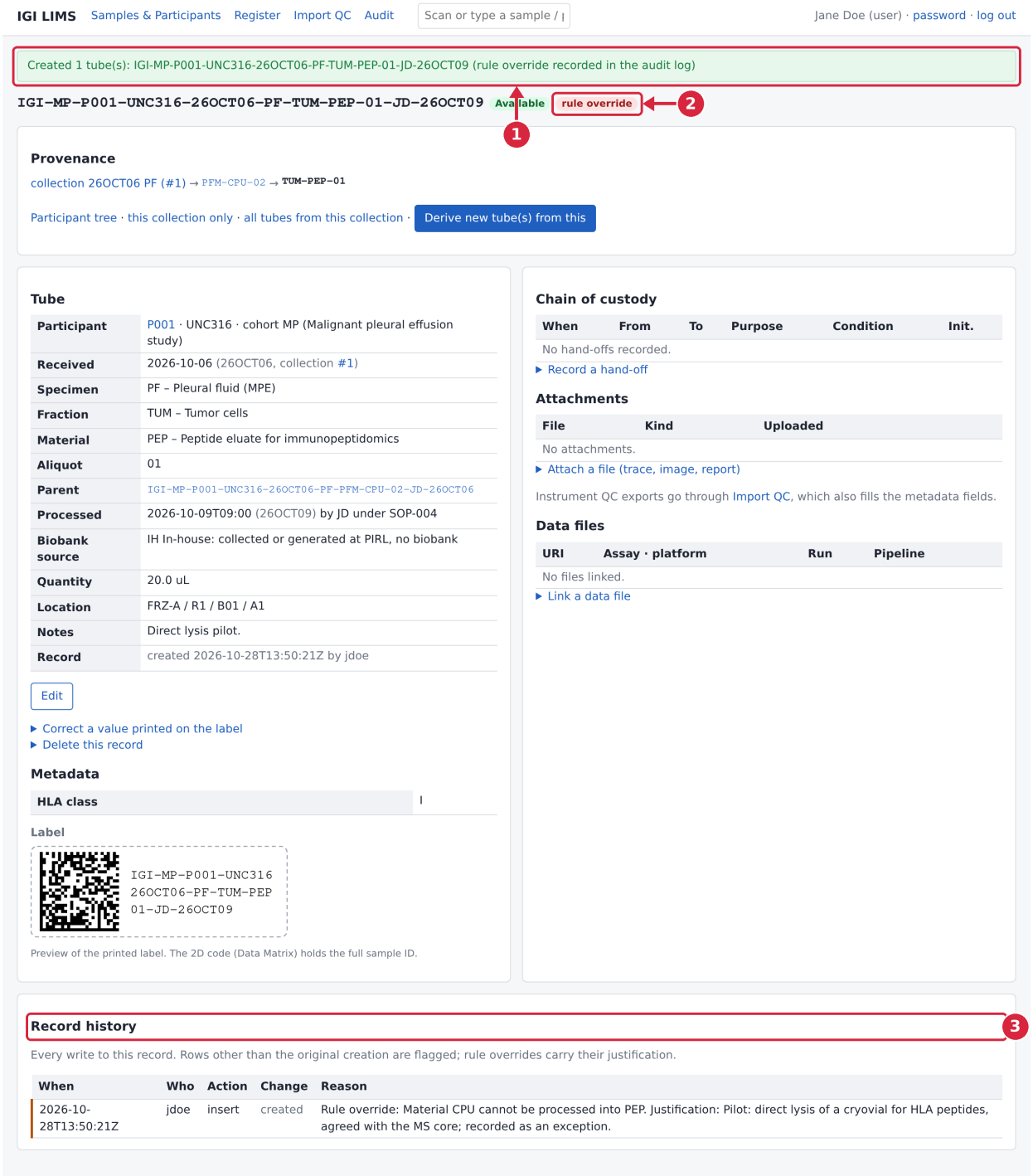
1 The rule table blocks this derivation; nothing was saved

2 The specific rule broken: cells must be pelleted first

3 Create anyway: needs a justification, and is flagged in the audit log

6.5 The tube created by the override

The tube created by the override.



1 Created, with the override noted

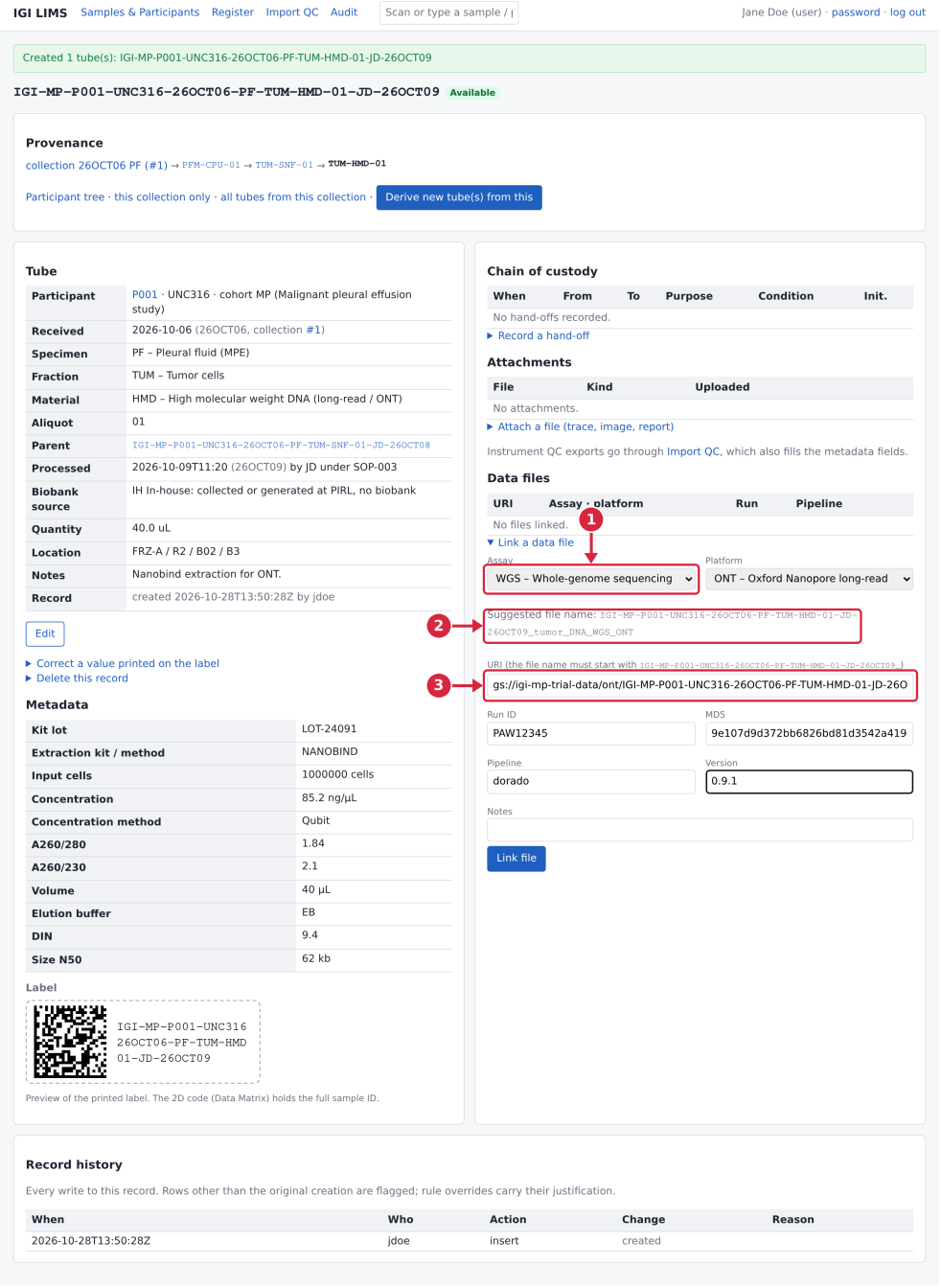
2 The tube carries a 'rule override' flag for life

3 The history row holds the rule broken and the justification

6.6 Link a data file

HMW DNA was then extracted from the first pellet (IGI-MP-P001-UNC316-26OCT06-PF-TUM-HMD-01-JD-26OCT09) and sequenced on ONT. The data file form requires an assay and a platform and suggests the file name the nomenclature asks for: the sample ID, then the source, assay and platform labels.

Linking a sequencing output to the HMW DNA tube handed to the facility.



1 Assay and platform are required lists

2 The suggested name: sample ID, source, assay, platform

3 Bucket URI; the file name must start with the sample ID

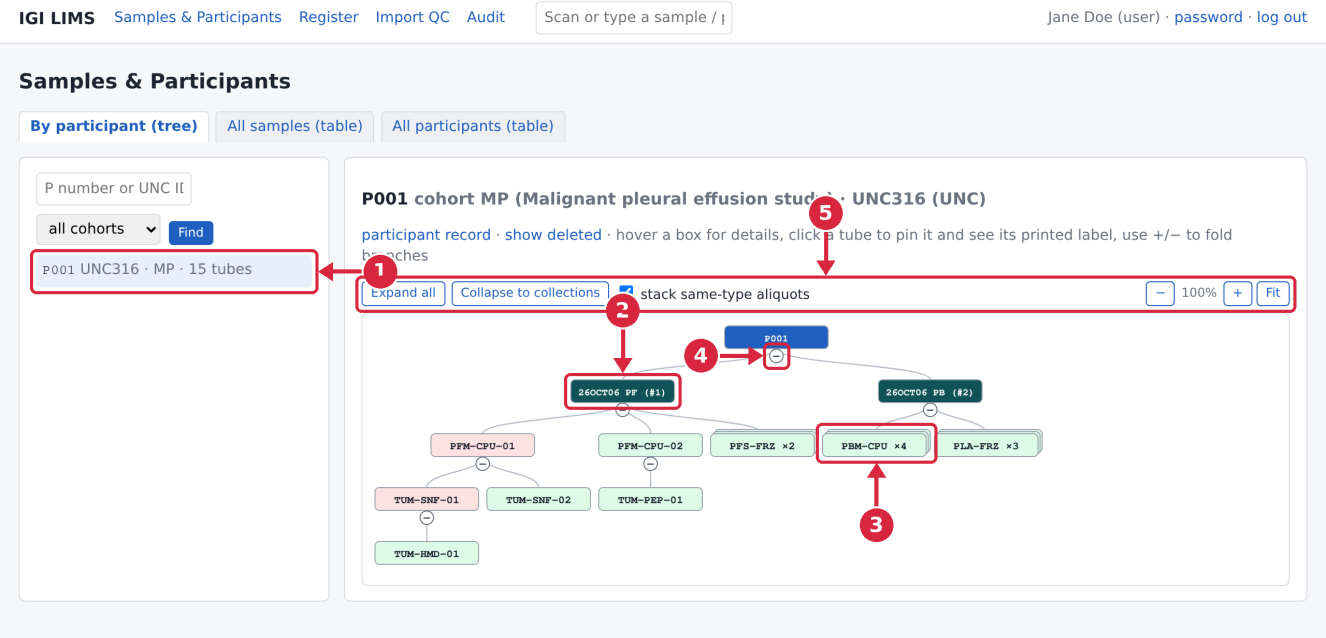
7. Walkthrough: browsing

Samples & Participants shows a participant's collections and everything made from them as an interactive tree. It is read-only, so it is safe to leave open on a shared screen.

7.1 The participant tree

Sibling aliquots of the same kind that have nothing made from them are stacked into one box, such as PBM-CPU ×4; clicking it spreads them out. Every branch can be folded with its +/– handle, and the toolbar expands everything, folds back to the collections, turns stacking off, or zooms. The tree fits the window until you zoom.

Samples & Participants: the participant's collections and everything derived from them, top down. Colours follow status.



1 Pick a participant

2 A collection (dark): the specimen as received

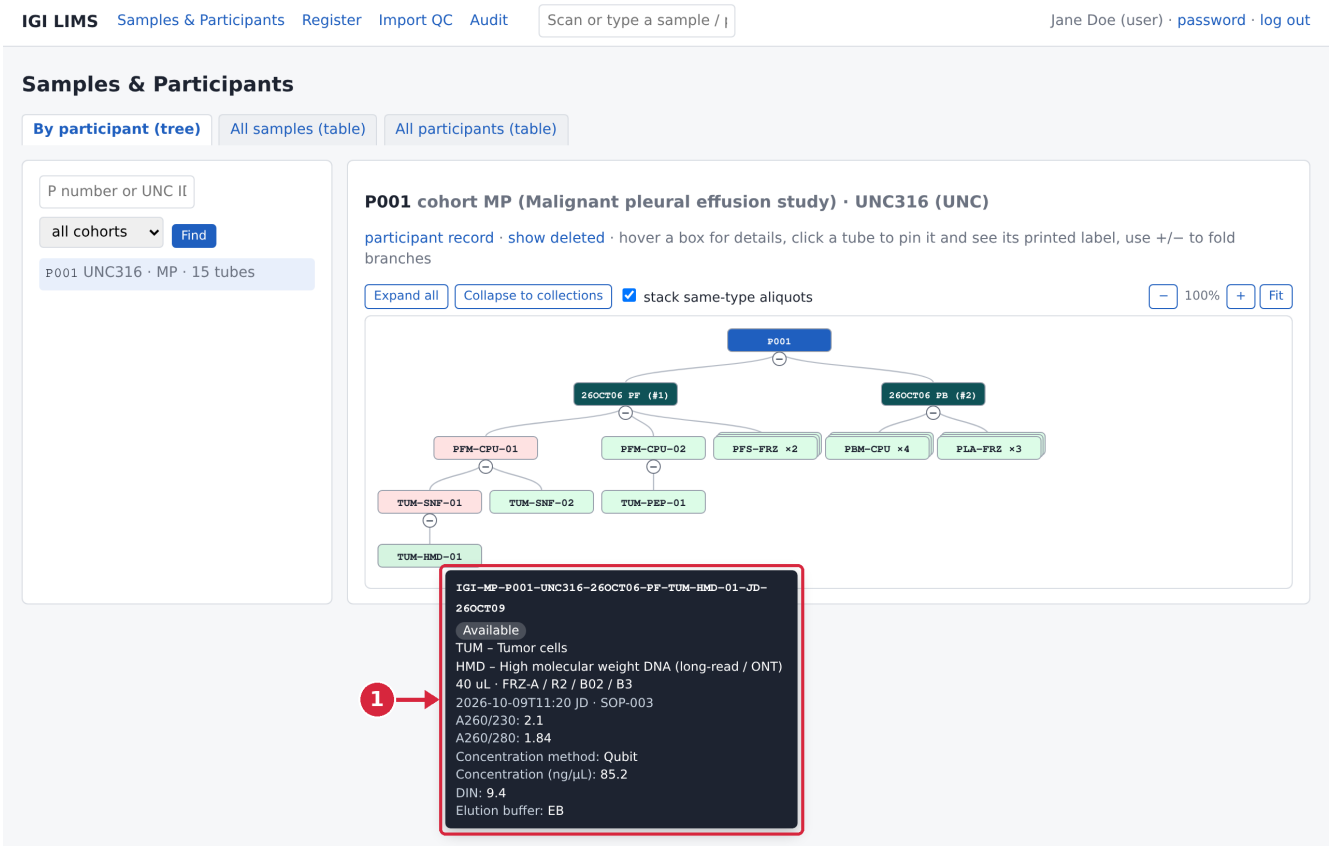
3 Stacked aliquots: click to spread them out

4 Fold or unfold any branch

5 Expand all, collapse to collections, stacking, zoom

7.2 Hover for details

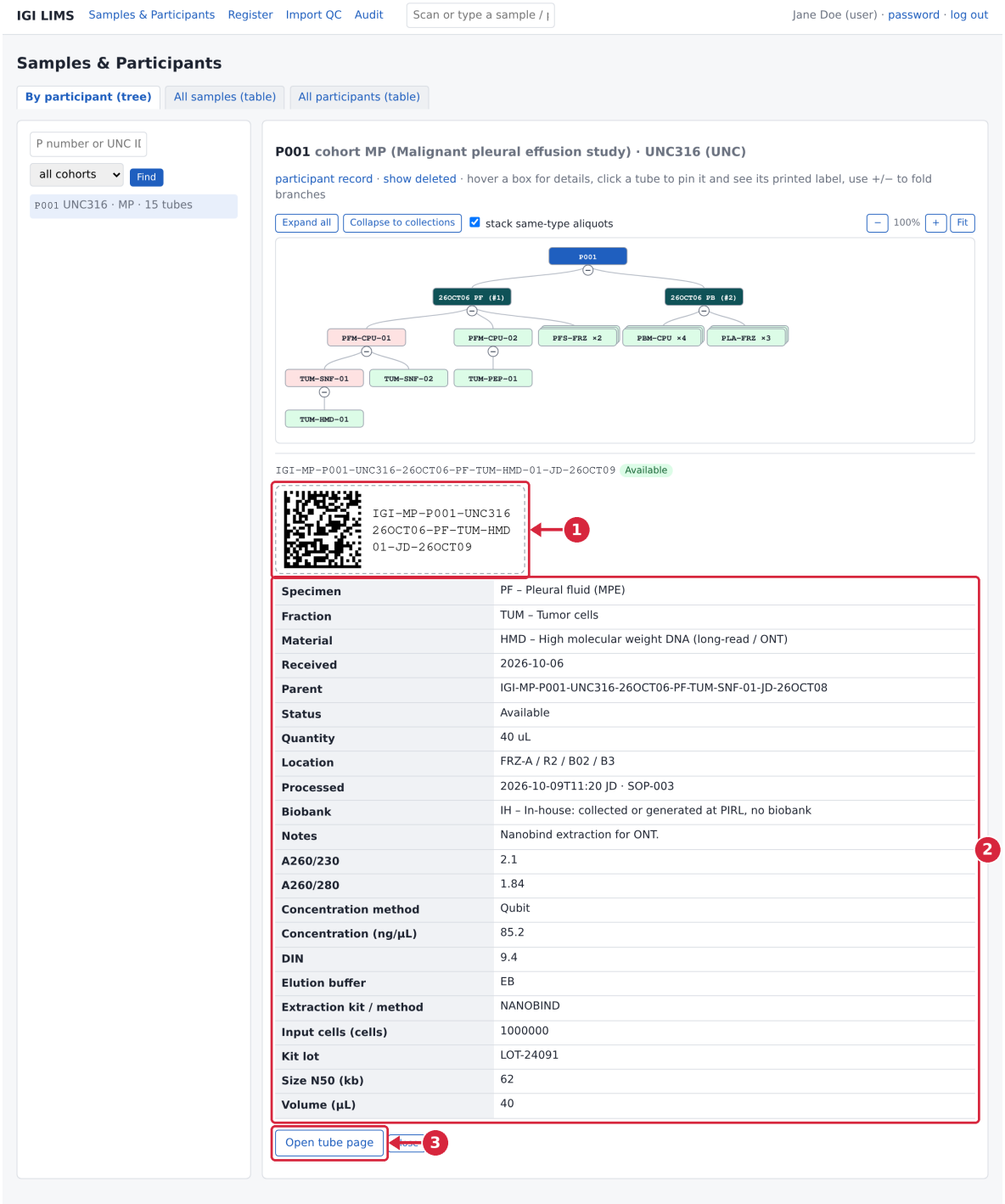
Hovering a tube.



1 Hover shows status, quantity, location and key metadata

7.3 Click to pin, with the label

Clicking a tube pins its details and its label.



1 The printed label with its 2D barcode

2 All fields and metadata

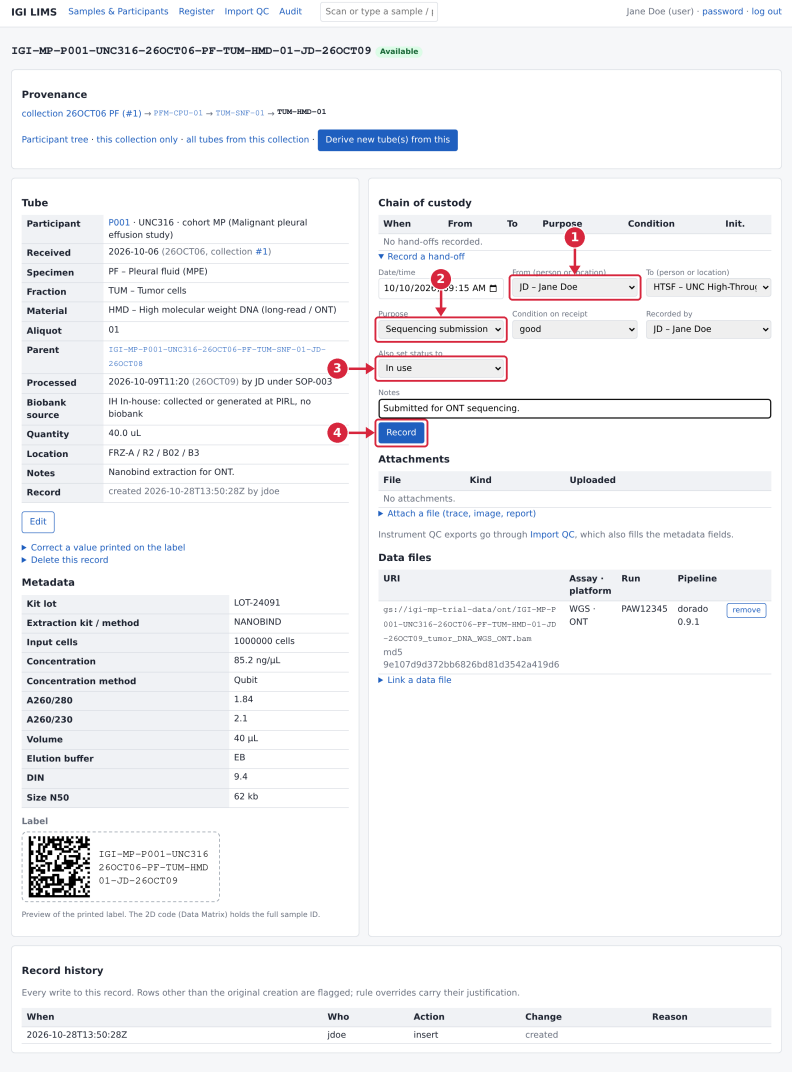
3 On to the tube page

8. Walkthrough: custody, editing and corrections

8.1 Record a hand-off

Custody events are append-only. Each records from, to, purpose, time and condition. The DNA tube is handed to the sequencing core here and marked In use in the same step.

Recording a hand-off of the DNA tube to the sequencing core.



1 From: a user or a party such as HTSF or CCPD

2 Purpose, from the custody purpose list

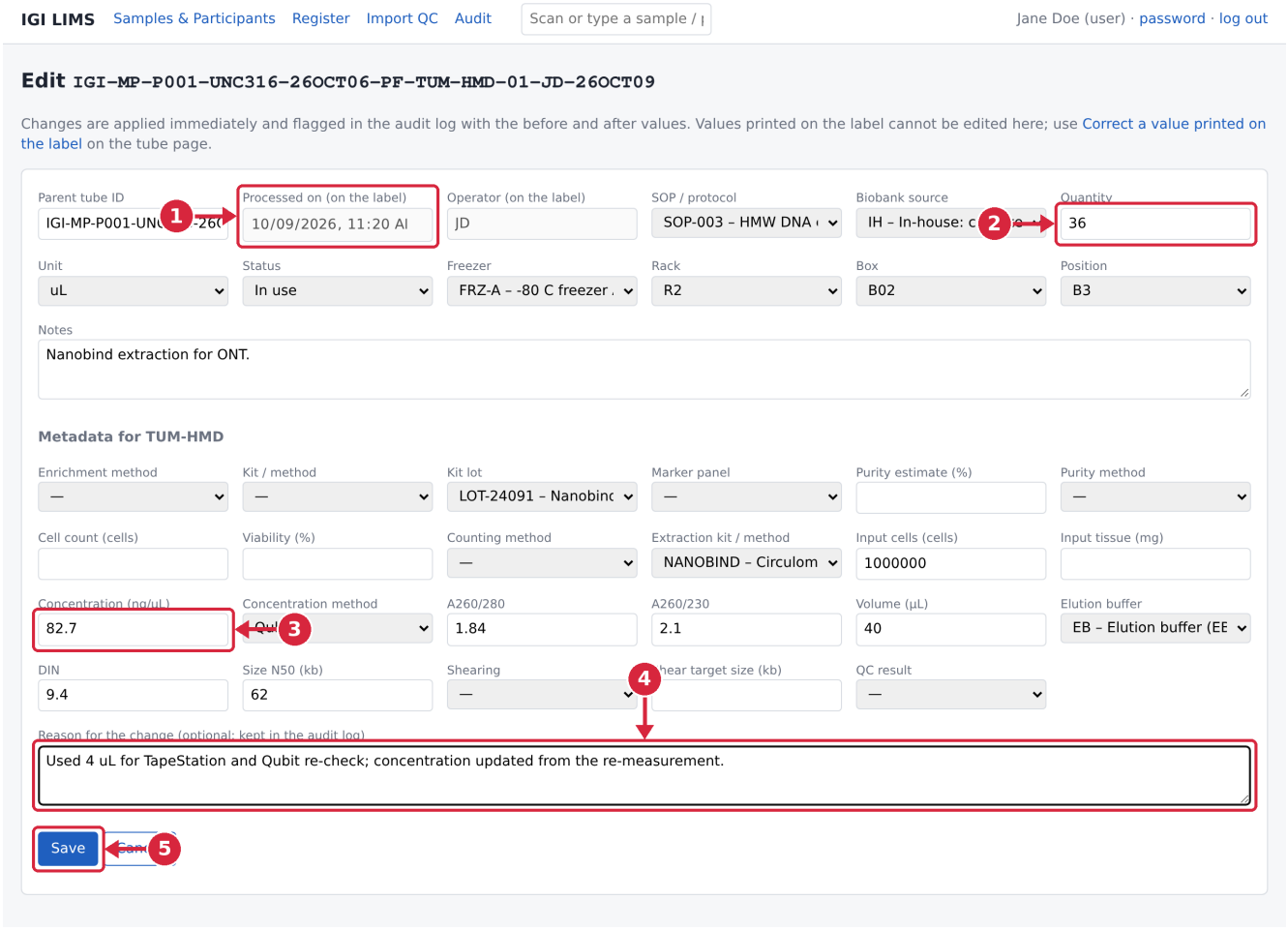
3 Optionally update the tube status in the same step

4 Record

8.2 Modify a record

Jane Doe re-measured the DNA and used some of it for QC, so the quantity and concentration change. Values printed on the label are shown but cannot be edited.

Modifying a record. Before and after values and the reason go to the audit log.

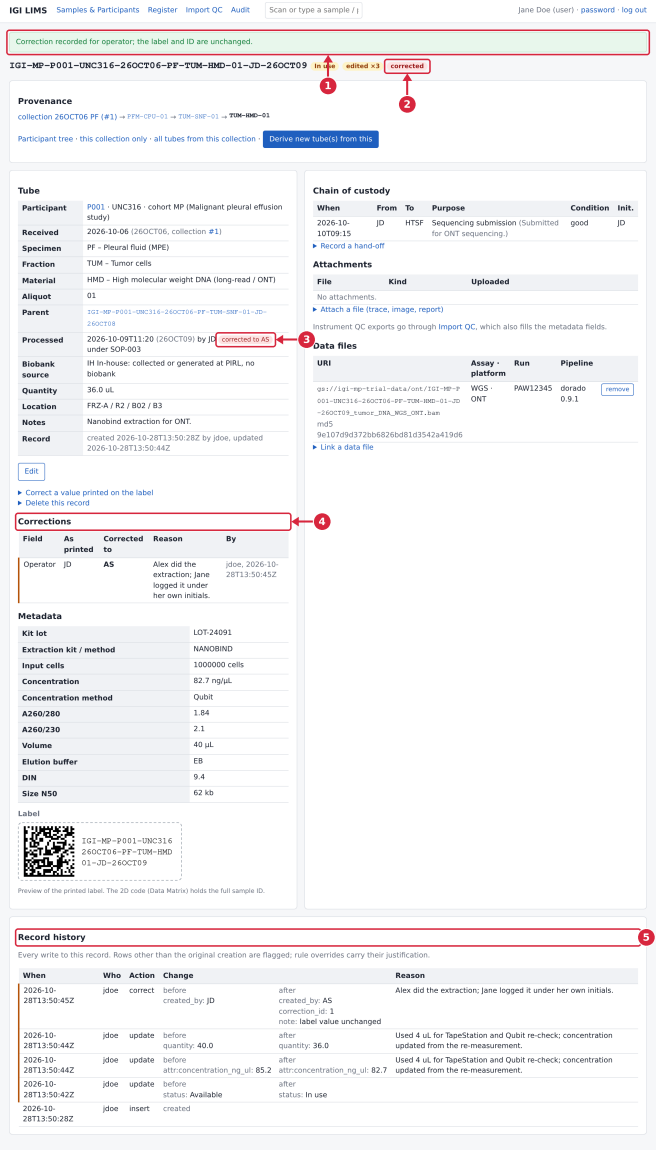


1. Printed on the label: read-only (corrected instead)
2. Quantity 40 to 36
3. Metadata is edited on the same form
4. Optional reason, kept with the audit row
5. Save

8.3 Correct a value printed on the label

The DNA was logged with Jane's initials, but Alex did the extraction. The label already says JD, so the operator is corrected rather than edited: the tube shows 'corrected to AS' beside the printed value, and the correction, the edit and the custody status change are all in the history. A second user, Alex Smith, then moves a cryovial to another box with an ordinary edit.

The DNA tube after an edit and a correction. The operator printed on the label (JD) was wrong; the correction records AS without rewriting the ID.



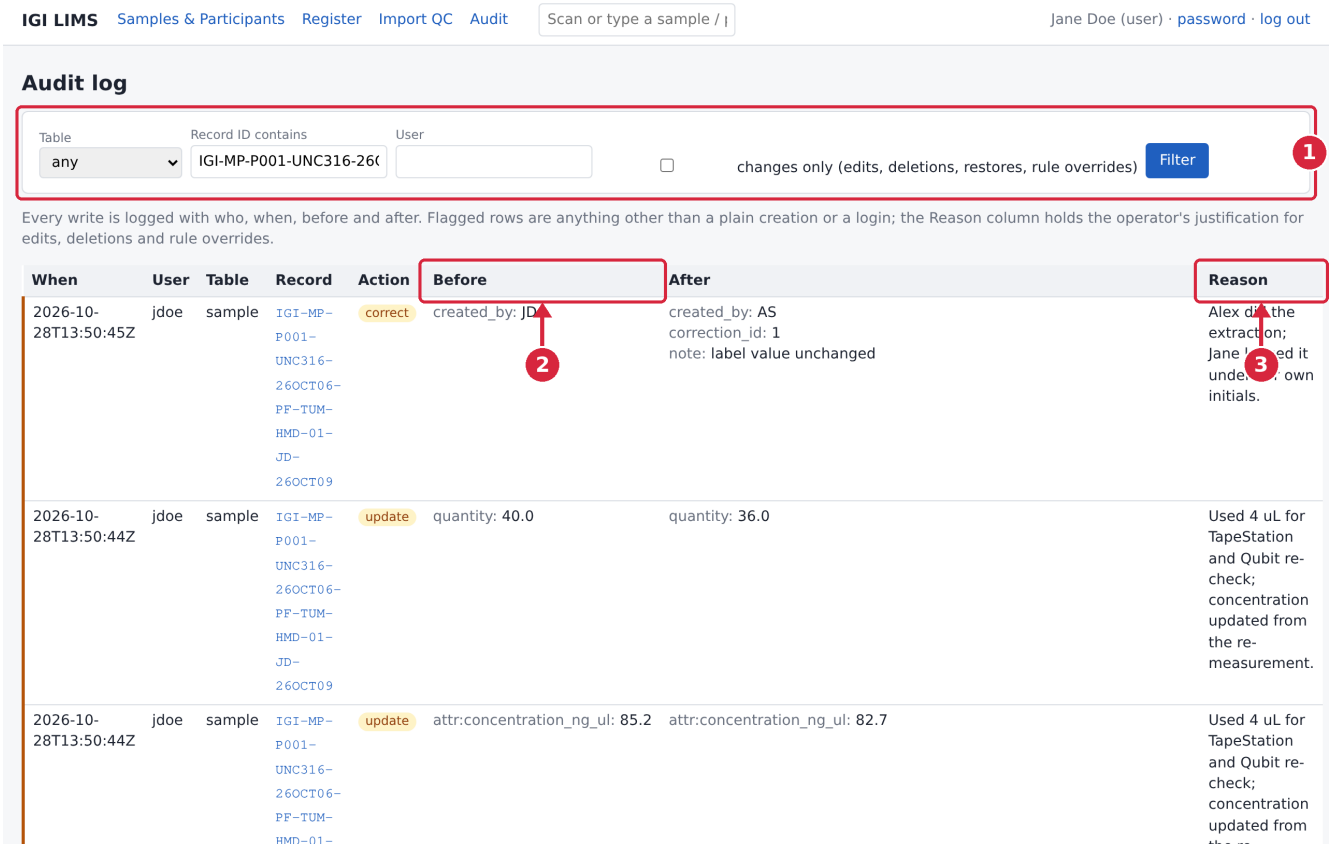
1. Correction recorded; label and ID unchanged
2. The tube is tagged as edited and corrected
3. The printed value stays; the correction sits beside it
4. Every correction with its reason and author
5. Edits, custody status change and correction, all flagged

9. Walkthrough: the audit log

9.1 Everything that happened to one tube

The audit log is the system-wide version of the history block on each tube page. It covers participants, tubes, custody, data files, metadata, attachments, code lists, fields and users. Before and After are shown as one key and value per line so a reviewer can read them without parsing JSON.

The audit log filtered to one tube. Creation, edits, custody and corrections all appear here; changes are flagged.



1 Filter by table, record ID or user, or show changes only

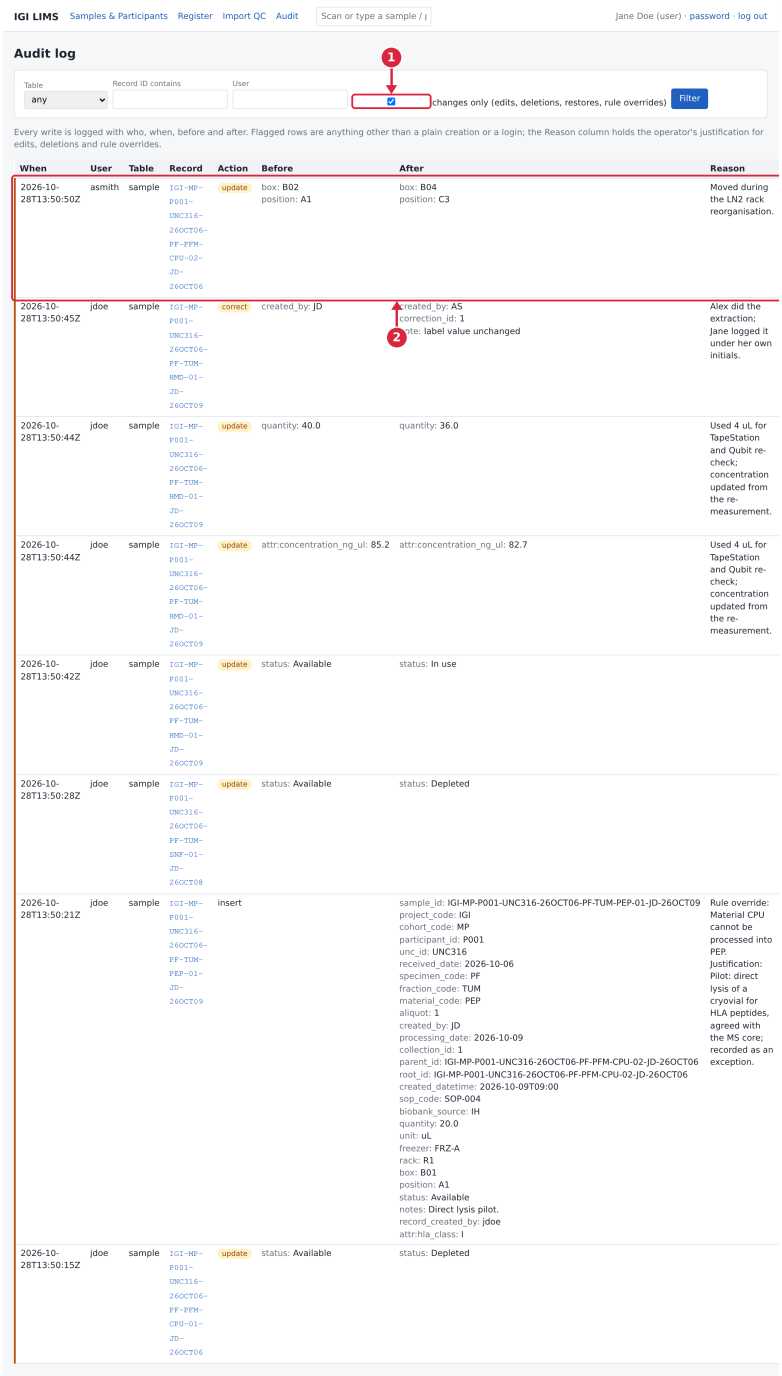
2 Before and After as key: value lines

3 The operator's stated reason for edits, deletions and overrides

9.2 Changes only

One checkbox hides routine creations and logins and leaves edits, corrections, deletions, restores and rule overrides, each with who, when, before, after and the stated reason.

The changes-only view. Alex Smith's vial move, Jane Doe's edit, the correction, the custody status change and the earlier rule override are all here.



1 Changes only: edits, deletions, restores and rule overrides

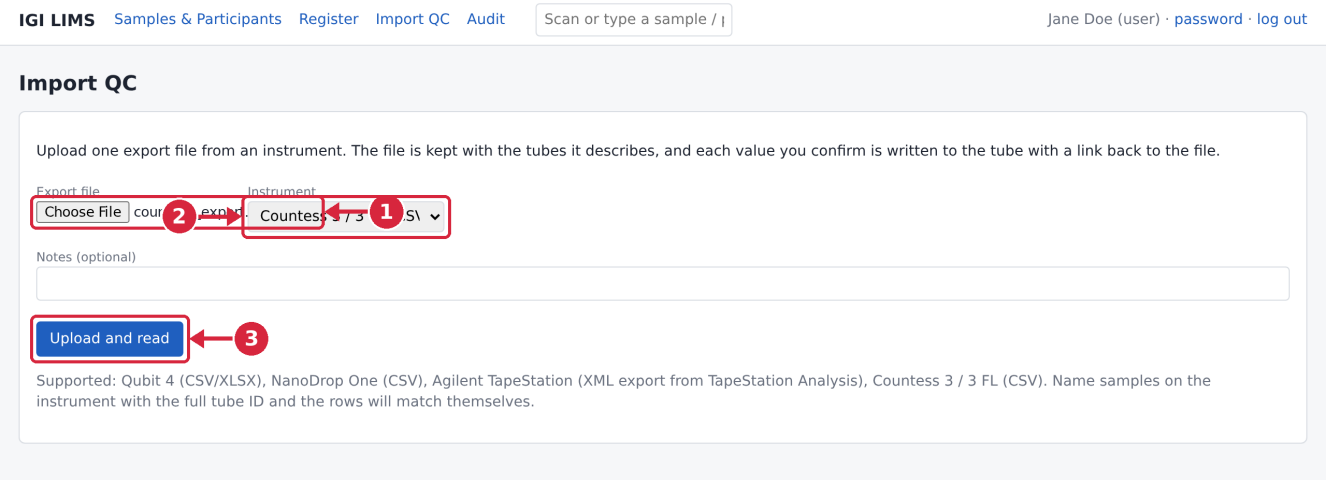
2 Each flagged row names who, when, what changed and why

10. Walkthrough: importing instrument QC files

Common instrument exports can be loaded directly instead of typing numbers into metadata fields. Supported today: Qubit 4 and NanoDrop One exports (parsed through the allotropy library), TapeStation XML, and Countess 3 CSV. The example below imports a Countess cell count.

10.1 Upload one export

Importing a Countess 3 cell-count export.



1 One instrument export file

2 Instrument: auto-detected if left blank

3 Upload and parse

10.2 Confirm what goes where

One row per instrument sample. Rows with no tube ID are skipped; the blank slide stays blank.



1 The instrument's sample name

2 Suggested tube ID: matched because the name contains it

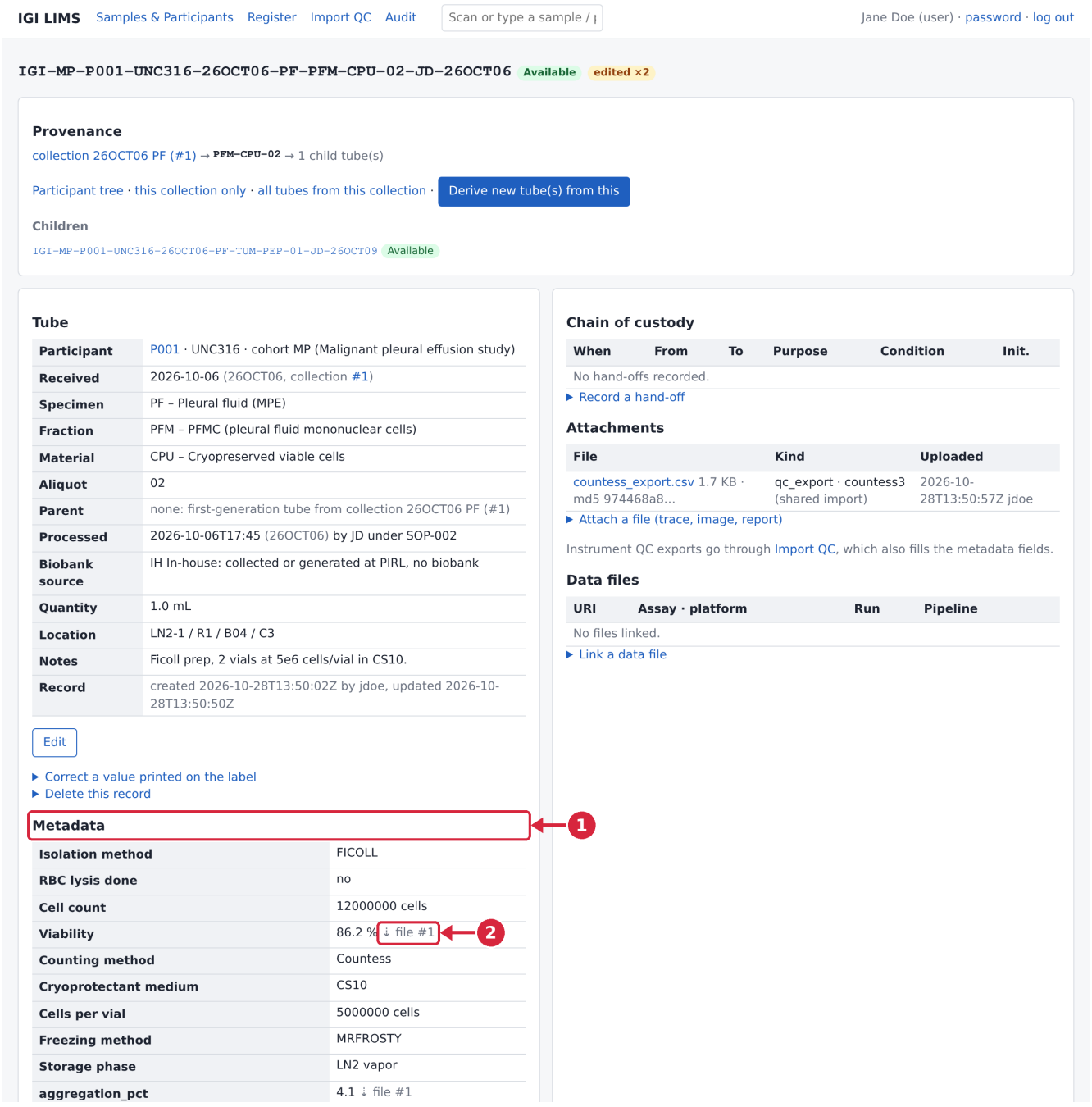
3 Untick any value you do not want written

4 Write values to tubes

10.3 Provenance of imported values

Each value written by an import keeps a link to the stored export file, so a concentration or viability can always be traced to the instrument run it came from. The export itself is kept as an attachment.

The second cryovial after import. Each imported value points back to the file it came from.



1 Imported values, with a link to the stored export file

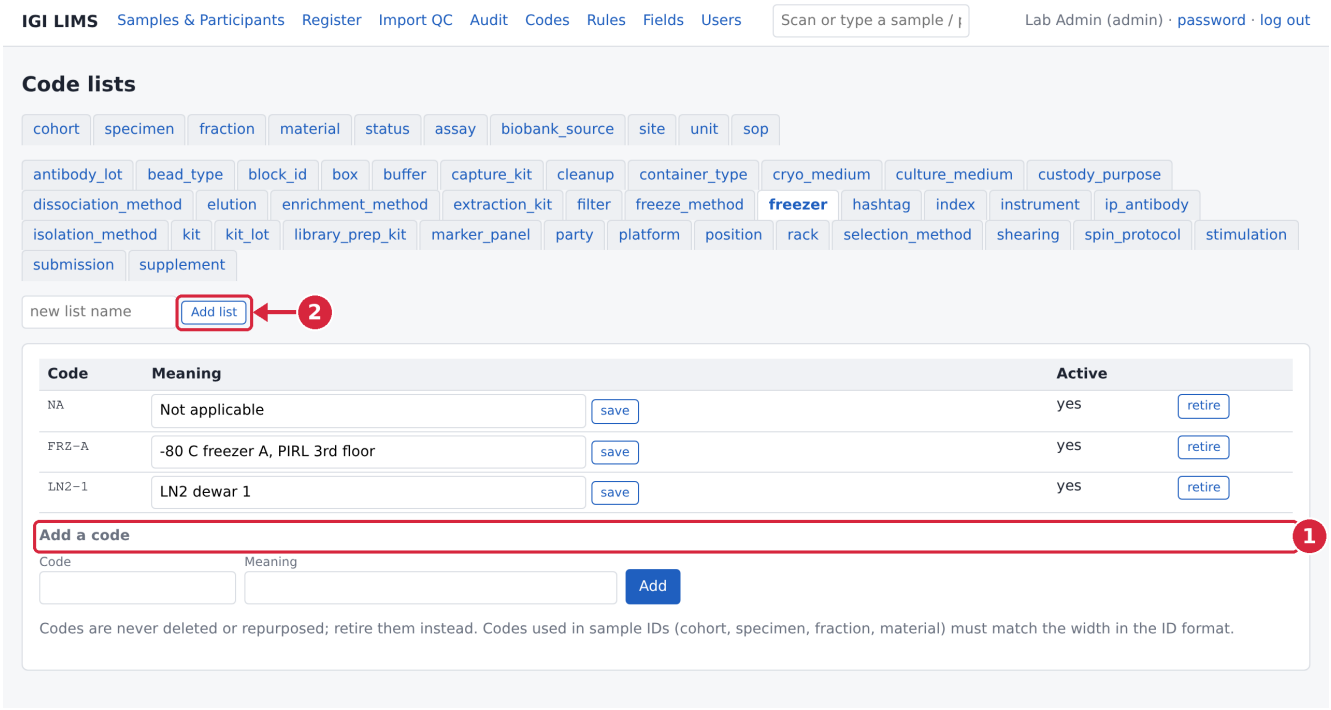
2 Source file

11. Administration

Administrators see four extra pages. None of them require a code change or a restart.

11.1 Codes: every controlled vocabulary

The Codes page. Entries are retired, never deleted, so old records keep their meaning. Codes used in IDs must match their segment width.



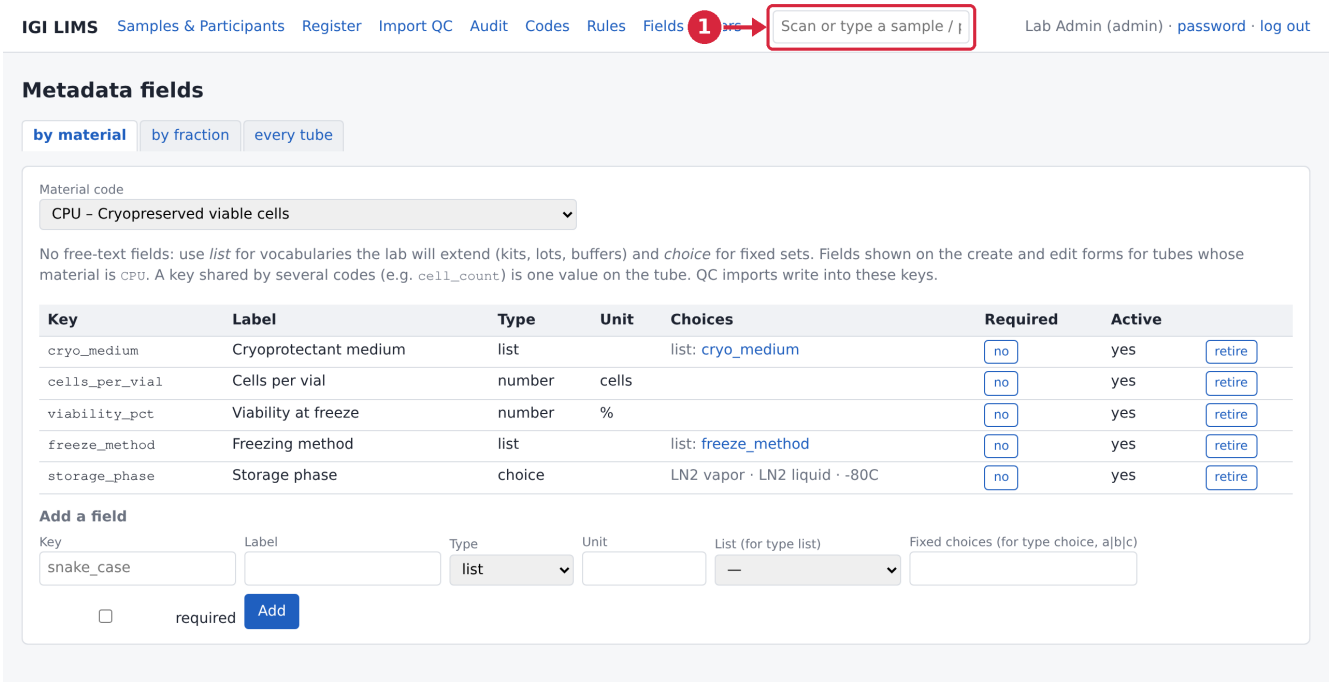
1 Add an entry to any list

2 Create a whole new list for a new metadata field

11.2 Fields: which metadata each fraction and material carries

Fields are typed as number (with unit), fixed choice, admin-extendable list, yes/no, date or date-time. There is no text type. A field can be marked required.

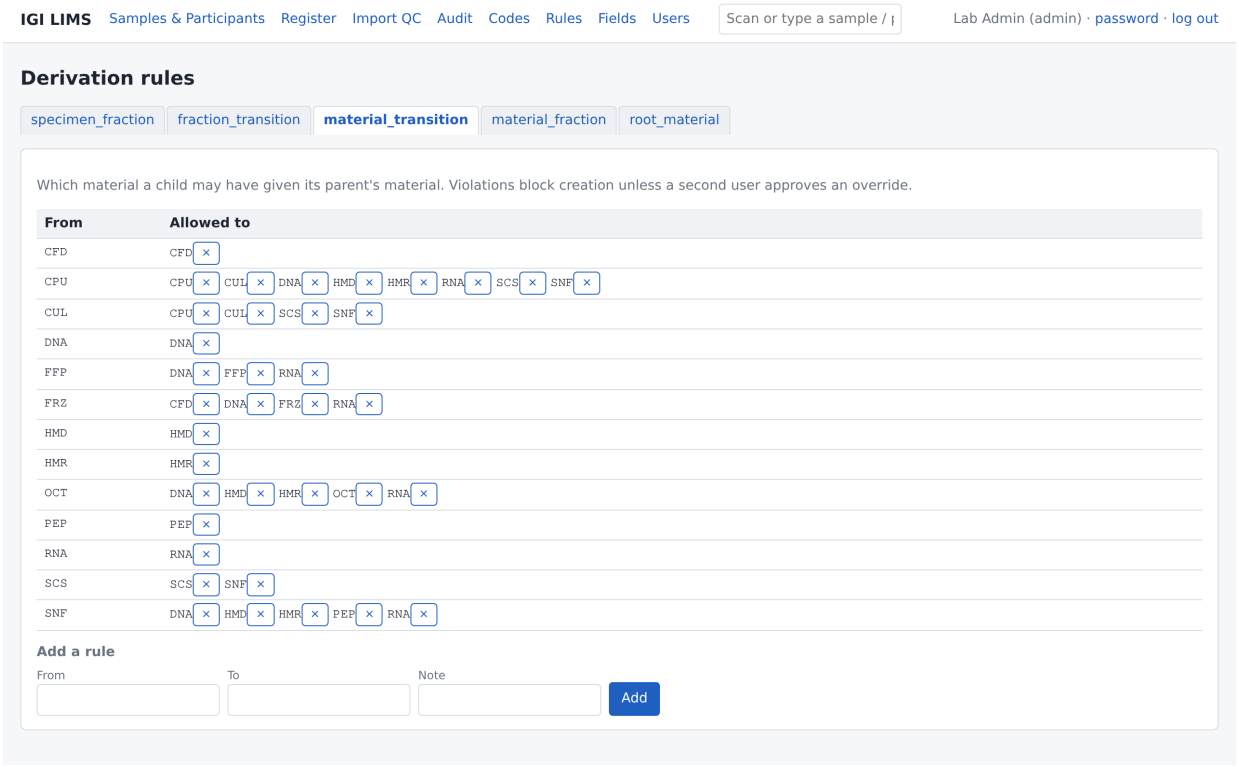
The Fields page defines which metadata each fraction or material carries.



1 Add a metadata field: scope, label, type, unit, list, required

11.3 Rules: allowed specimens, fractions and transitions

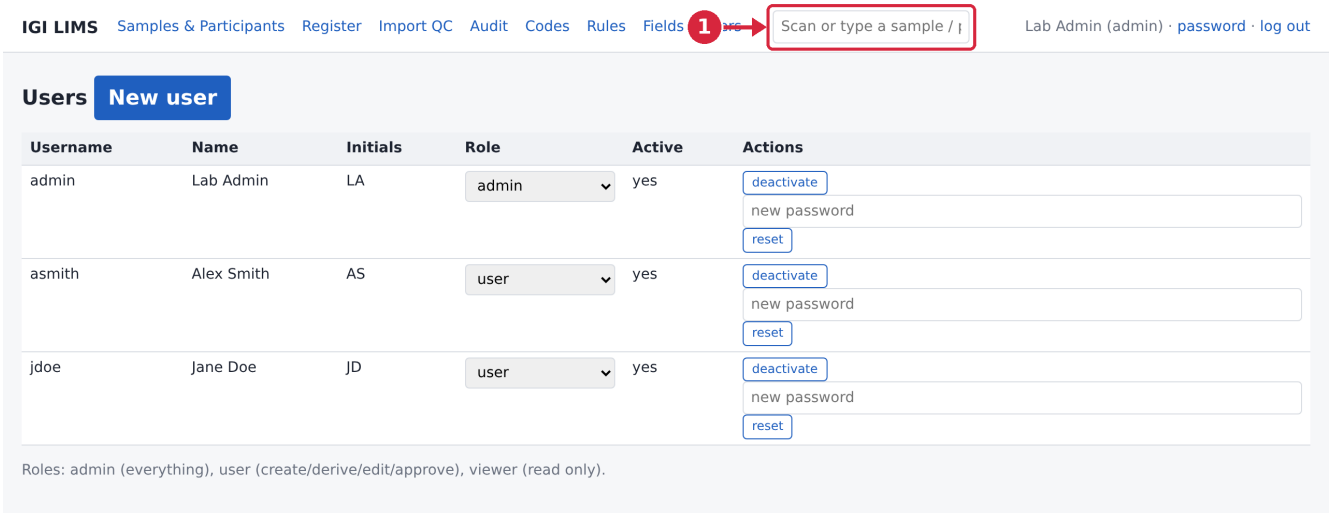
The Rules page: which fractions a specimen may have, and which transitions are allowed.



11.4 Users and roles

Roles are admin (everything), user (create, derive, edit) and viewer (read only).

The Users page.



1 Create a user with a role: admin, user or viewer

12. Also available

Data files

Sequencing and mass-spectrometry outputs live in Google Cloud Storage buckets. A data file record links a bucket URI to the tube handed to the facility (usually DNA, HMD, RNA, HMR or a peptide eluate), with assay, platform, run ID, checksum and pipeline version, so a file is always reachable from the specimen it came from. File names start with the sample ID and an underscore, for example `…-HMD-01-KEN-26OCT06_tumor_DNA_WGS_ONT.bam`.

Attachments

Any file (trace, image, report) can be attached to a tube. It is stored on disk under the attachment directory with an md5 checksum and listed on the tube page.

Read-only JSON API

For pipelines and scripts. Authenticate with a browser session or an `X-Api-Token` header matching the token set in the server environment.

Endpoint	Returns
/api/pattern	anchored regex for IDs
/api/validate/<id>	the ID parsed into its eleven components, and whether it exists
/api/samples?participant=&collection=&material=&status=&prefix=	list of tubes
/api/samples/<id>	one tube with its collection, custody events, files and corrections
/api/samples/<id>/lineage	ancestors and all descendants
/api/tree/<id>	nested tree of the tube's collection
/api/collections/<collection_id>	a collection with its tree and corrections
/api/files	data files with their tube and lineage

Technical summary

- Python 3.10, Flask 3, Jinja templates, a small amount of vanilla JavaScript for the tree and dynamic forms.
- SQLite (one file, WAL mode) for local use, or PostgreSQL (Supabase) when hosted on Vercel; the same code runs on both.
- Tables: code_list, derivation_rule, account, participant, collection, sample, sample_correction, custody_event, data_file, audit_log, attribute_definition, sample_attribute, attachment.
- Instrument parsing via allotropy (Benchling open source) for Qubit and NanoDrop; TapeStation and Countess parsed directly.
- 66 automated tests cover ID composition, the nomenclature's worked example, collections, registration, editing, corrections, overrides, parsers, import, metadata and vocabulary enforcement; they run against SQLite and PostgreSQL.
- Hosting: a public demonstration instance on Vercel with Supabase Postgres (docs/DEPLOY_VERCEL_SUPABASE.md); a GCP VM remains an option for production.

13. Open questions (tracked in NOTES.md)

- The v2.1 nomenclature document is still under team review; the build follows the 2026-10-19 text.
- File-name labels for bulk TCR and immunopeptidomics, and whether 10x V(D)J needs its own platform label.
- Two collections of the same specimen received on the same day share the first six ID segments.
- IRB and HIPAA review of a label carrying the patient ID, the UNC ID and two dates.
- The seeded derivation rules are a proposal and need review by the lab.
- Which metadata fields should be required.
- The Countess parser follows the vendor column specification and has not yet been checked against a real export.